

Journal Pre-proof

Multimomics Reveals Biological Mechanisms Linking Macroscale Structural Covariance Network Dysfunction With Neuropsychiatric Symptoms Across the Alzheimer's Disease Continuum

Jiwei Jiang, Kun Zhao, Wenyi Li, Peiyang Zheng, Shirui Jiang, Qiwei Ren, Yunyun Duan, Huiying Yu, Xiaopeng Kang, Junjie Li, Ke Hu, Tianlin Jiang, Min Zhao, Linlin Wang, Shiyi Yang, Huiying Zhang, Yaou Liu, Anxin Wang, Yong Liu, Jun Xu

PII: S0006-3223(24)01666-4

DOI: <https://doi.org/10.1016/j.biopsych.2024.08.027>

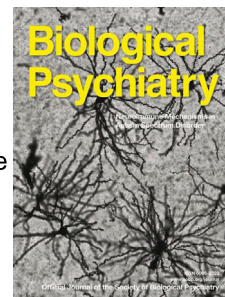
Reference: BPS 15626

To appear in: *Biological Psychiatry*

Received Date: 1 February 2024

Revised Date: 4 July 2024

Accepted Date: 28 August 2024



Please cite this article as: Jiang J., Zhao K., Li W., Zheng P., Jiang S., Ren Q., Duan Y., Yu H., Kang X., Li J., Hu K., Jiang T., Zhao M., Wang L., Yang S., Zhang H., Liu Y., Wang A., Liu Y. & Xu J., Multimomics Reveals Biological Mechanisms Linking Macroscale Structural Covariance Network Dysfunction With Neuropsychiatric Symptoms Across the Alzheimer's Disease Continuum, *Biological Psychiatry* (2024), doi: <https://doi.org/10.1016/j.biopsych.2024.08.027>.

This is a PDF file of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability, but it is not yet the definitive version of record. This version will undergo additional copyediting, typesetting and review before it is published in its final form, but we are providing this version to give early visibility of the article. Please note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2024 Published by Elsevier Inc on behalf of Society of Biological Psychiatry.

1 **Multimomics Reveals Biological Mechanisms Linking Macroscale**
2 **Structural Covariance Network Dysfunction With Neuropsychiatric**
3 **Symptoms Across the Alzheimer's Disease Continuum**

4 Running Title: Structural Covariance Network Unraveling NPS in AD

5 Jiwei Jiang ^{a,b}, Kun Zhao ^{c*}, Wenyi Li ^{a,b}, Peiyang Zheng ^c, Shirui Jiang ^{a,b}, Qiwei Ren ^{a,b},
6 Yunyun Duan ^a, Huiying Yu ^d, Xiaopeng Kang ^e, Junjie Li ^a, Ke Hu ^e, Tianlin Jiang ^{a,b}, Min
7 Zhao ^{a,b}, Linlin Wang ^{a,b}, Shiyi Yang ^{a,b}, Huiying Zhang ^{a,b}, Yaou Liu ^a, Anxin Wang ^{a,b},
8 Yong Liu ^{c*}, Jun Xu ^{a,b*}

9

10 ^a Beijing Tiantan Hospital, Capital Medical University, Beijing, 100070, China

11 ^b China National Clinical Research Center for Neurological Diseases, Beijing 100070,
12 China

13 ^c School of Artificial Intelligence, Beijing University of Posts and Telecommunications,
14 Beijing, 100876, China

15 ^d School of Information Science and Engineering, Shandong Normal University, Jinan,
16 251100, China

17 ^e School of Artificial Intelligence, University of Chinese Academy of Sciences, Beijing
18 100876, China

19

20 *Correspondence:

21 Dr. Jun Xu

22 Department of Neurology, Beijing Tiantan Hospital, Capital Medical University; China

23 National Clinical Research Center for Neurological Diseases, Beijing 100071, China

24 Email: neurojun@126.com

25

26 Dr. Yong Liu

27 School of Artificial Intelligence, Beijing University of Posts and Telecommunications,
28 Beijing 100876, China
29 Email: yongliu@bupt.edu.cn

30

31 Dr. Kun Zhao

32 School of Artificial Intelligence, Beijing University of Posts and Telecommunications,
33 Beijing 100876, China
34 Email: kunzhao@bupt.edu.cn

35

36 Abstract

37 Background: The highly heterogeneity of neuropsychiatric symptoms (NPSs) hinder
38 further exploration of their role in neurobiological mechanisms and Alzheimer's
39 disease (AD). We aimed to delineate NPS patterns based on brain macroscale
40 connectomics to understand the biological mechanisms of NPSs on the AD
41 continuum.

42 Methods: We constructed Regional Radiomics Similarity Networks (R2SN) for 550
43 participants (AD with NPSs [AD-NPS, n=376], AD without NPSs [AD-nNPS, n=111], and
44 normal controls [n=63]) from CIBL study. We identified R2SN connections associated
45 with NPSs, and then cluster distinct subtypes of AD-NPS. An independent dataset
46 (n=189) and internal validation were performed to assess the robustness of the NPS
47 subtypes. Subsequent multiomics analysis were performed to assess the distinct
48 clinical phenotype and biological mechanisms in each NPS subtype.

49 Results: AD-NPS patients were clustered into severe (n=187), moderate (n=87), and
50 mild NPS (n=102) subtypes, each exhibiting distinct brain network dysfunction
51 patterns. A high level of consistency in clustering NPS was internally and externally
52 validated. Severe and moderate NPSs showed significant cognitive impairment,
53 increased plasma p-Tau₁₈₁ levels, extensive decreased brain volume and cortical
54 thickness, and accelerated cognitive decline. Gene set enrichment analysis (GSEA)
55 revealed enrichment of differentially expressed genes in ion transport and synaptic
56 transmission with variations for each NPS subtype. Genome-wide association studies
57 (GWAS) analysis defined the specific gene loci for each subtype of AD-NPS (i.e, logical
58 memory), aligning with clinical manifestations and progression patterns.

59 Conclusions: This study identified and validated three distinct NPS subtypes,
60 underscoring the role of NPSs in neurobiological mechanisms and progression of the
61 AD continuum.

62

63 Keywords: Alzheimer's disease; neuropsychiatric symptom; structural connectomics;
64 multimodal imaging; multiomics; validation

65

66

Journal Pre-proof

67 Introduction

68 Neuropsychiatric symptoms (NPSs) are common and important clinical
69 manifestations across the Alzheimer's disease (AD) continuum, characterized by
70 intricate and heterogeneous processes (1). Approximately 90% of patients experience
71 one or more neuropsychiatric disturbances during AD progression (2), which
72 contributes to overall deterioration in individuals with AD and increases caregiver
73 burden (3, 4). Despite their high prevalence and profound impact, NPSs are not part
74 of any diagnostic paradigm for AD, and the interrelationship between NPSs and AD
75 pathophysiological mechanisms remains incompletely elucidated (5). Recent research
76 has revealed that NPSs are prevalent across all clinical AD stages, but are not
77 associated with cognitive function over time at any clinical stage (6). The complexity
78 of NPS impedes our understanding of their role in AD and thus is adverse to
79 evidence-based intervention measures. Therefore, it necessitates clustering NPSs into
80 meaningful symptom groups with specific features.

81 A fundamental assumption in neuroscience is that the brain network connections
82 determine the brain's functional organization (7, 8). Neuroimaging connections in the
83 living brain provide an opportunity to identify the driving factors behind the
84 neurobiology of cognition and behavior (9). Recent advances in neuroimaging
85 methods have revealed that pathological proteins accumulate along specific
86 macroscale brain networks, implicating the brain network architecture in the
87 system-level pathophysiology of neurodegenerative diseases (10). Brain network
88 abnormalities could provide in vivo signatures for distinguishing proteinopathies
89 across and within these neurodegenerative diseases (11). The evidence provides new
90 insight into elucidating the complexity and heterogeneity of NPSs on the AD
91 continuum by macroscale brain connectome.

92 Brain structure observations reveal patterns of co-variation recognized as

structural covariance networks (SCNs) (12), which have recently emerged as a promising methodology for amalgamating various structural neuroimaging features into biologically meaningful individual-specific connectomes (13). Regional Radiomics Similarity Network (R2SN) is a novel and stable network that can help understand human morphological covariance and explore the biological mechanisms based on structural neuroimaging (14). It has been used successfully to investigate the dysfunction of brain networks and reveal potential biological characteristics in AD-related dementia (14-18).

Therefore, in this study, we aimed to investigate the R2SN connections associated with NPSs on the AD continuum and developed a data-driven subtyping framework to delineate distinct subtypes of AD patients with NPSs. We internally and externally validated the stability and repeatability of clustering NPS subtypes. Subsequently, we explored AD multiomics characteristics, including clinical presentations, plasma pathological biomarkers, multimodal neuroimaging burdens, and clinical progression, within these NPS subtypes. Further, we assessed the biological mechanisms underpinning the various NPS subtypes and their connections to the biological pathways leading to the AD continuum.

Methods and Materials

Study design and participants

We included patients on the AD continuum with NPSs (AD-NPS, n=376) and without NPSs (AD-nNPS, n=111), and normal controls (NCs, n=63) from the Chinese Imaging, Biomarkers, and Lifestyle (CIBL) study conducted between January 1, 2021 and February 28, 2023. The CIBL study, as previously described ([19](#)), was approved by the Ethics Committee of Capital Medical University, Beijing Tiantan Hospital.

We included patients with AD continuum including MCI and AD dementia (age of onset ≥ 60 years); the former diagnosed using Petersen's diagnostic criteria for amnesic MCI or the 2018 National Institute on Aging Alzheimer's Association (NIA-AA) workgroup diagnostic criteria for MCI due to AD and the latter using the 2011 or 2018 NIA-AA workgroup diagnostic criteria for probable AD or confirmed AD ([20](#), [21](#)). Among them, 168 (44.92%) patients had pathological biomarker findings meeting the AT(N) (amyloid/tau/neurodegeneration) framework including 27 CSF biomarkers, 6 PET/CT, and 135 plasma biomarkers.

All NCs were ≥ 50 years old, cognitively normal, lacking any general or laboratory evidence of other diseases that could impact cognition, and without any neuropsychiatric symptom. Detailed inclusion and exclusion criteria are in Supplementary Material S01. The study design and data analysis are shown in Figure 1.

Clinical data collection and assessments

Clinical data on demographics and clinical characteristics, including age, sex, years of education, marital status, medical history, and apolipoprotein E4 allele (*APOE* ϵ 4) status, were collected. Comprehensive neuropsychological assessments, including the Mini-Mental State Examination (MMSE), Montreal Cognitive Assessment (MoCA), Neuropsychiatric Inventory (NPI), Activities of Daily Living (ADL), Mini-Nutritional

Assessment (MNA), Caregiver Burden Inventory (CBI), and Pittsburgh Sleep Quality Index (PSQI), were also performed. The content and criteria of each neuropsychological scale are detailed in Supplementary Material S02.

Plasma pathological biomarkers

In total 135 patients on the AD continuum were detected with plasma pathological biomarkers. The plasma concentrations of the 40 and 42 amino acid-long amyloid- β ($A\beta$) fragments ($A\beta_{40}$ and $A\beta_{42}$), tau phosphorylated at amino acid 181 (p-Tau₁₈₁), and total tau (t-Tau) were measured using BD FACSCanto II flow cytometry. The measurements were performed in one round of experiments using one batch of reagents, with the analysts blinded to diagnosis and clinical data.

Multimodal neuroimaging data acquisition and analysis

All participants underwent structural three-dimensional (3D) T1-weighted and 7-delay pseudo-continuous arterial spin labeling (pCASL) image acquisition using a 3.0-T magnetic resonance (MR) scanner (SIGNA Premier; GE Healthcare, Milwaukee, WI, USA) with a 48-channel head coil. The open-source Computational Anatomy Toolbox v12.8 (CAT12; <http://www.neuro.uni-jena.de/cat/>) was used to segment each participant's T1-weighted image and calculate the regional gray matter (GM) volume and cortical thickness (CT) (22) according to the Brainnetome Atlas (23). Additionally, regional corrected cerebral blood flow (cCBF) was extracted from the preprocessed pCASL image (24). Considering the lower spatial resolution of cCBF, the cCBF value was extracted using the third version of the automated anatomical atlas (178 brain regions) (25). The multimodal neuroimaging parameters and calculated processes are detailed in Supplementary Material S03.

Individualized R2SN construction

To comprehensively access the brain connectome patterns, we introduced a novel stable morphological covariation network, entitled R2SN, which has been used

successfully to reflect connectome changes in the brain based on structural T1-weighted MR imaging data (14, 15, 26). The individualized R2SN was created using an open-access toolkit at https://github.com/YongLiuLab/R2SN_construction (14). Each participant's 3D T1-weighted MR image was first aligned to the Montreal Neurological Institute space using Advanced Normalization Tools (<https://github.com/ANTsX/ANTs>). A series of radiomics features were extracted for 246 regions defined by the Brainnetome Atlas (27). Here, 22 radiomics features were extracted for each brain region, details presented in Supplementary Material S04. Finally, the R2SN was calculated using Pearson's correlations between interregional normalized radiomics features (14).

Relationship between the dysfunction in R2SN profiles and NPSs

To explore the relationship between dysfunction in R2SN profiles and NPSs, we conducted a two-sample, two-sided t-test comparing patients with AD-NPS and AD-nNPS. Connections showing significant differences (if $P < 0.01$) between the two groups were considered closely related to NPSs. Furthermore, we employed a clustering method called non-negative matrix factorization (NMF) (28) to categorize the NPS subtypes based on the aforementioned dysfunction in R2SN profiles (the connections showed significant differences between AD-NPS and AD-nNPS) among AD-NPS patients. This framework for subtyping analysis was widely used in previous studies (26, 29). The optimal number of clusters for patients with AD-NPS was determined by analyzing the silhouette coefficient's value and changes in the residual sum of squares (29). The detailed NMF process and silhouette coefficient calculation are presented in Supplementary Material S05.

Distinct multiomics measures among distinct subtypes of NPSs with specific dysfunctions of the R2SN

We first computed dysfunctions in the R2SN profiles in each NPS subtype in the AD

continuum by comparing them to those of patients with AD-nNPS via a two-sample, two-sided t-test, respectively. To characterize the distinct clinical manifestations among different dysfunctions of the R2SN in NPSs, we comprehensively explored the difference in demographic characteristics, clinical presentations, plasma pathological biomarkers, neuroimaging burden, neuropsychological assessments and clinical progression between each NPS subtype and patients with AD-nNPS.

To evaluate the stability and reliability of the clustering findings, 10000 cycles of fully random permutation tests were performed for all comparisons in patients in the AD-NPS group. Importantly, we conducted internal and external validations to confirm the robustness and generalization of our clustering methods. Detailed can be found in Supplementary Material S06.

The biological mechanism for distinct dysfunction of R2SN in patients with AD-NPS

We performed gene set enrichment analysis (GSEA) to compute the potential relationship between gene expression and anatomical abnormalities for each NPS subtype. Gene datasets based on the group-level transcriptome analysis were sourced from the Allen Human Brain Atlas (AHBA) (<https://human.brain-map.org/>) (30). This resource provides approximately 500 samples per hemisphere, covering the cerebrum, cerebellum, and brainstem, each profiled with 62000 gene probes. Using the Abagen toolkit (31), we mapped all gene data onto the Brainnetome Atlas. Finally, 15633 genes from 236 brain regions were obtained. For this analysis, the T-map of altered regional GM in various NPS subtypes compared to AD-nNPS served as an independent variable with gene expression as the dependent variable. Following the previous study (32), the partial least squares (PLS) analysis was used to estimate the gene expressions most strongly correlated with the altered regional GM and to unveil the biological mechanisms of each NPS subtype based on these genes using the Metascape platform (<https://metascape.org/gp/index.html#/main/step1>)

(33). To further investigate the specific biological underpinnings of distinct dysfunctions in the R2SN among patients with AD-NPS, genome-wide association studies (GWAS) were conducted for all participants on the AD continuum. Details can be found in Supplementary Material S07.

Longitudinal progression of the clinical measure for each NPS subtype than those AD-nNPS patients

Generalized linear mixed-effects models with a random intercept and slope were utilized to discern the longitudinal association between each NPS subtype and subsequent change in the neuropsychological assessments in patients on the AD continuum during the follow-up period. The ordinary least squares (OLS) regression was employed to identify the association between the follow-up time and neuropsychological assessments of each patient. The age, sex, education, and *APOEε4* status were considered confounding factors.

229 Results

230 Baseline demographic and clinical characteristics of all participants

231 Table 1 presents the clinical and biomarker characteristics of all participants at
 232 baseline. Patients within the AD continuum were generally older, had a lower
 233 proportion of females, lower educational levels, and higher frequencies of *APOE*ε4
 234 carriers, as well as a medical history of hypertension and cerebrovascular disease,
 235 compared to NCs (all $P < 0.05$). However, no significant difference was observed in
 236 these conditions and plasma pathological biomarkers between patients in the
 237 AD-NPS group and the AD-nNPS group (all $P > 0.05$). Patients in the AD-NPS group had
 238 more serious neuropsychological assessments than those in the AD-nNPS group (all
 239 $P < 0.05$). The NPI scores in the AD-NPS group were 8.00 (4.00, 15.00).

240 R2SN profiles showed significant differences among NCs, AD-nNPS, and AD-NPS

241 Substantial R2SN connections differed significantly among the NCs, AD-nNPS, and
 242 AD-NPS groups (Supplementary Figure S1). The altered R2SN connections in the
 243 AD-NPS group were primarily located in the psychobehavioral-related brain areas,
 244 including the precentral gyrus, thalamus, superior frontal gyrus, and cingulate gyrus
 245 (Figure 2A). However, the altered R2SN connections in the AD-nNPS group were
 246 located primarily in the regions related to cognition: the hippocampus, superior
 247 parietal lobule, and precuneus (Figure 2B). Furthermore, 300 key connections were
 248 highly related with NPSs, significantly different between the two groups ($P < 0.01$), e.g.,
 249 the thalamus, dorsolateral putamen, superior frontal gyrus, inferior temporal gyrus,
 250 and insular gyrus (Figure 2C).

251 Data-driven method characterized the distinct dysfunction of R2SN in AD-NPS

252 Comparative analyses with AD-nNPS revealed complex alterations in R2SN among AD
 253 patients with three different NPS subtypes. Specifically, the first NPS subtype ($n=187$)

exhibited significant modifications in the medioventral occipital cortex, inferior temporal gyrus, fusiform gyrus, and thalamus (Figure 2D), whereas the second NPS subtype (n=102) showed minimal differences compared to AD-nNPS (Figure 2E). Conversely, the third NPS subtype (n=87) displayed an altered pattern opposite to the first subtype (Figure 2F). Compared to the second subtype, the other two subtypes demonstrated significant changes in the morphological connectivity of the paracentral lobules, insular gyrus, postcentral gyrus, and parietal lobule (Figure 2G–I). Additionally, we performed clustering analyses using the top 100 and top 200 features that demonstrated significant differences between AD-NPS and AD-nNPS. Consequently, the NPS subtypes exhibited high consistency across different connections (84.1% and 88.6% were overlapped). The detailed findings are shown in Supplementary Table S1 and Table S2.

The clustering results were approximately reproducible between different datasets and brain atlas

In the internal validation, the mean correlation coefficients consistently exceeded 0.4, with 82.9% surpassing 0.7. Additionally, the maximum correlation coefficient in each resampling exceeded 0.7 in 92.2% of cases, while the minimum correlation coefficient exceeded 0.7 in 69.3% of cases, demonstrating the good stability of the clustering method (Supplementary Figure S2A–C).

Patients in the validated datasets showed lower MMSE and MoCA scores, but higher ADL and CBI scores than those in the original datasets (all $P < 0.05$; Supplementary Table S3). High consistency for the centroids of clusters was shown in the first NPS subtype ($R = 0.70$, $P < 0.001$) and the second subtype ($R = 0.58$, $P < 0.001$) between the original and validated datasets. However, the centroids of clusters for the last NPS subtype demonstrated significant yet weaker consistency ($R = 0.18$, $P = 0.002$) between the two datasets (Supplementary Figure S2D–F).

Meanwhile, the first (75%) and second (61%) NPS subtypes demonstrated high consistency between the R2SN models based on different atlases, while the last NPS subtype data points were clustered into the first and second NPS categories (Supplementary Table S4). Overall, the first two NPS subtypes demonstrated high robustness across different datasets and atlases, while the last NPS subtype exhibited weak consistency, reflecting its role as a transitory stage between the first and second subtypes. Consequently, some data points were categorized into first and second NPS groups (Supplementary Table S5).

Distinct dysfunction of R2SN is related to the different clinical phenotypes. Supplementary Table S6 shows the clinical and biomarkers differences between each NPS subtype and the AD-nNPS, respectively. Compared with AD-nNPS, the first and last subtypes exhibited serious performance on the neuropsychological assessments (all $P < 0.05$). Conversely, no significant difference was found in the MMSE and MoCA scores between the second NPS subtype and AD-nNPS (both $P > 0.05$). Similarly, compared to those in the AD-nNPS group, patients with the first and last subtypes showed significantly increased plasma p-Tau₁₈₁, while no significant difference in these plasma biomarkers was seen in patients with the second subtype.

The distinctions in 12 neuropsychiatric characteristics among three subtypes of NPSs are delineated in Supplementary Figure S3. Patients with the first NPS subtype exhibited a heightened frequency of agitation and irritability than those with the other two subtypes (all $P < 0.05$).

As three subtypes exhibited distinct dysfunction of the R2SN, different clinical phenotypes, and validated findings, we defined them as severe NPS (sNPS), mild NPS (miNPS), and moderate NPS (moNPS). The permutation test yielded $P < 0.05$ for all comparisons among the three NPS subtypes after 10,000 permutations.

Distinct dysfunction of the R2SN is related to the multimodal neuroimaging burden

Compared to AD-nNPS, both sNPS and moNPS subtypes exhibited reduced regional GM volume in the temporal lobe, hippocampus, nucleus accumbens, and thalamus (Figure 3A and C), and widespread reductions in the regional CT across the frontal, temporal, and parietal lobules (Figure 3D and F). Conversely, no significant difference was observed in the regional GM volume (Figure 3B) or CT between patients with miNPS and AD-nNPS (Figure 3E). Despite the absence of significant structural brain changes, patients with miNPS exhibited higher cCBF values in the inferior frontal gyrus, putamen, and anterior cingulate cortex than AD-nNPS (Figure 3H), suggesting early compensatory cerebral hyperperfusion. A comparative analysis of the multimodal neuroimaging burden of the three NPS subtypes is presented in Supplementary Figure S4.

Neurobiological mechanism of the anatomic variability among the three distinct NPS subtypes based on transcriptome and GWAS

The *T*-maps of the regional GM volume of the three NPS subtypes versus AD-nNPS served as a neuroimaging measure of the anatomical variability of each subtype. Significant correlations were observed between the first PLS component (PLS1) and the anatomic variability of the three subtypes (Figure 4A–C). The GSEA based on transcriptome unveiled dysfunctional ion transmembrane transport and synaptic signaling across all three subtypes, but the types substantially varied. Patients with sNPS (Figure 4D) showed significant enrichment in delayed rectifier potassium channel activity ($P_{\text{FDR}}=8.128\times10^{-9}$), voltage-gated sodium channel activity ($P_{\text{FDR}}=2.884\times10^{-4}$), regulation of catecholamine secretion ($P_{\text{FDR}}=7.244\times10^{-4}$), and dopaminergic synapse pathway ($P_{\text{FDR}}=2.691\times10^{-6}$). Conversely, patients with moNPS (Figure 4F) exhibited enrichment in cytosolic calcium ion concentration ($P_{\text{FDR}}=1.820\times10^{-4}$), calcium ion binding ($P_{\text{FDR}}=3.311\times10^{-4}$), ligand-gated monoatomic anion channel activity ($P_{\text{FDR}}=6.607\times10^{-3}$), monoatomic ion channel complex ($P_{\text{FDR}}=3.467\times10^{-4}$), glutamatergic ($P_{\text{FDR}}=6.607\times10^{-4}$) and GABAergic synapses

($P_{FDR}=0.017$). Notably, patients with miNPS did not exhibit significant neurotransmitter changes, and enrichment was observed in only the Gene Ontology terms related to biological processes (Figure 4E).

After quality controls and imputation, 997,369 SNPs were included in the GWAS. We identified two genomic risk loci significantly associated with sNPS, with the top SNPs mapping to 13 genes in these regions. These genes are enriched in the pathway related to logical memory in mild cognitive impairment, aligning with clinical manifestations and progression patterns (Figure 5A). Additionally, we identified five genomic risk loci significantly associated with miNPS, with the top SNPs in these regions mapping to 38 genes enriched in the type 2 diabetes pathway (Figure 5B). Furthermore, we discovered five genomic risk loci significantly associated with moNPS, enriched in the pathway for non-response to antidepressants and depression (Figure 5C).

Longitudinal clinical progression in the three subtypes of NPS

Among 487 patients on the AD continuum at baseline, 165 (34.52%) completed follow-up (60 sNPS, 49 miNPS, 27 moNPS), with an average duration of 10.56 months. The longitudinal association between the three NPS subtypes at baseline and clinical progression is presented in Supplementary Figure S5. Comparative analysis revealed that patients with sNPS and moNPS at baseline exhibited a significant decline in MMSE and MoCA scores over the follow-up period after adjustment (all $P<0.05$). However, no significant differences in longitudinal changes were observed in MMSE and MoCA scores between patients with miNPS and AD-nNPS ($P=0.173$, $P=0.143$).

356 Discussion

357 In this study, we systematically identified and validated three distinct subtypes of
358 NPSs on the AD continuum using a data-driven methodology, which helped address
359 their subjectivity and heterogeneity of clinical measurements. We explored brain
360 connectome differences in individuals with NPSs and gained a better understanding
361 of their contribution to brain connectivity, cognitive progression, neuroimaging, and
362 gene expression on the AD continuum. These findings could potentially promote the
363 development of tailored treatment approaches for patients with NPSs on the AD
364 continuum.

365 We identified three NPS subtypes with distinct brain connection damage patterns,
366 offering a step forward in NPS management and precise interventions for AD patients.
367 Heterogeneous network patterns provide crucial information on neural signaling (34).
368 A previous study identified two different MCI subtypes by employing the R2SN,
369 thereby providing new insights for risk assessment and precise early intervention in
370 these patients (27). These suggest brain connections analysis (such as R2SN) could
371 identify shared and distinct mechanisms of complex diseases. Our study found
372 patients with sNPS and moNPS had more serious cognitive impairment and rapid
373 deterioration on the AD continuum, whereas patients with miNPS did not show
374 significant differences. This indicates that the distinct dysfunctions of the R2SN are
375 associated with different cognitive presentations and outcomes. Consequently, the
376 R2SN could be considered a bridge between NPSs and AD neuropathology and
377 prognosis.

378 Individuals with NPSs had an inconsistent trajectory of changes in cognitive
379 function (6); moreover, a previous study showed that no significant correlation was
380 observed between NPSs and plasma pathological biomarkers across the entire AD
381 continuum (35), causing difficulties in clinical treatment and management (36).

However, another large cohort study found that AD patients with and without NPSs reported similar cognitive performance, and the level of NPS severity was independent of the diagnosis (37). Our study identified three distinct NPS subtypes and highlighted the potential association of different NPS subtype with plasma p-Tau₁₈₁ levels (which has been well confirmed to be related to the cognition (38, 39), cognitive impairment and progression. More importantly, the functional enrichment analysis of GWAS illustrated that sNPS subtype associated variants were mainly enriched in the logical memory, consistent with our clinical findings. Thus, this study helps divide the highly heterogeneous NPSs into three distinct subtypes related to cognition within the AD continuum, contributing to enhanced disease risk stratification and the development of potential precision medicine approaches for patients with NPSs.

Our study regarding multimodal neuroimaging found that patients with sNPS showed a decline in GM volume in the temporal lobe, whereas those with moNPS showed decreased GM volume in the frontal lobe, aligned with the opposite patterns of brain connection injury. Consistent with our findings, an observational study identified two canonical patterns between the GM structure and NPSs: the first pattern reflected a smaller volume in the adjacent temporal regions, whereas the second pattern reflected a smaller volume in the lateral and orbital prefrontal regions compared with the NCs (40). Our study found patients with miNPS showed no significant differences in brain structural measures than AD-NPS, but showed higher cerebral perfusion in the inferior frontal gyrus, putamen, and anterior cingulate cortex, indicating early-stage compensatory cerebral hyperperfusion (41). These functional changes were also consistent with the mild clinical cognitive impairment and progression of this subtype.

Over 40 genetic risk loci have been identified for AD (42); however, causal variants, regulatory elements, genes, and pathways of NPSs in the AD continuum remain

primarily unknown (43). The extent to which the differences among the three NPS
 subtypes are genetically driven requires further exploration. In line with previous
 evidence (15, 26, 44), our study used GSEA based on GWAS and transcriptome
 demonstrated the inorganic ion transmembrane transport (sodium and potassium)
 and synaptic signaling (catecholamine and dopamine) were correlated with sNPS,
 whereas calcium and anion transmembrane transport and synaptic signaling
 (glutamate and GABA) were correlated with moNPS, suggesting different
 neurobiological mechanisms between these two subtypes. A gene analysis study
 found 16 co-expressed gene modules enriched in ion transmembrane transport were
 associated with pathological hallmarks (neurofibrillary tangles) and cognition in AD
 development (45). Another study found that decreased astrocyte calcium signaling
 accompanied brain network activity and functional connectivity disruptions, leading
 to NPSs in early AD (46). The causative dependence of AD on ion channel mutations
 is unknown, but studies have hypothesized that different ion channels are distributed
 in different axonal subdomains to regulate neural network activity by initiation,
 conduction, and synaptic transmission (47). Mounting evidence suggests that brain
 neurotransmitter deficits and dysfunction can modulate a broad spectrum of
 neuropsychiatric diseases (48, 49). A recent study found that potassium increased
 dopamine uptake through the *Drosophila* dopamine transporter in liposomes and
 visualized Na⁺ and potassium fluxes in single proteoliposomes using fluorescent ion
 indicators (50), consistent with our study's findings on sNPS. However, a recent male
 C57BL/6 mouse study demonstrated that calcium-sensing receptor (CaSR) activation
 in the posttrauma area inhibited acute feeding, and R568 (a CaSR agonist) increased
 the glutamate and GABA content in the posttrauma area (51), suggesting a
 connection between calcium receptors and GABA, similar to our moNPS findings. To
 some extent, our study explained the differences and relationships between the
 three NPS subtypes and disease prognosis at the molecular and cellular levels. This

potentially benefits for understanding the difference between the clinical-radiological and R2SN characteristics and developing individualized NPS treatments.

Despite these contributions, this study has several limitations. First, the GSEA results are tentative, given that they largely depend on the genetic data derived from only six healthy subjects in the Allen Brain dataset. Moreover, GWAS analysis results are limited to the study's sample size. Second, due to the expensiveness and invasiveness of CSF testing and positron emission tomography, not all patients on the AD continuum in this study were diagnosed by the ATN framework. Future research should involve more comprehensive independent datasets to test the robustness of this study's findings. As working in native space is often considered the optimal procedure when working with variables defined based on atlases across subjects, thereby an R2SN based on native space should be further considered.

Conclusions

We identified three distinct NPS subtypes on the AD continuum through a data-driven approach. Our study elucidated the similarities and differences among different NPS subtypes in granular individual-level detail, and explored their contributions to AD neurobiological mechanisms and progression using multiomics and multimodal neuroimaging analysis. Further research should be conducted to validate the effectiveness and benefits of NPS management based on our data-driven NPS subtypes within the AD continuum.

Acknowledgments

The authors thank all patients and their caregivers for their participation and the personnel for contributing to the CIBL study.

460 Disclosures

461 Ethics approval and consent to participate

462 The Chinese Imaging, Biomarkers, and Lifestyle study received approval from the
463 Ethics Committee of Capital Medical University, Beijing Tiantan Hospital (approval
464 number: KY-2021-028-01). All participants or their legally authorized caregivers
465 provided written informed consent after receiving information about the study's
466 purpose.

467 Consent for publication

468 All participants or their legally authorized caregivers had signed informed written
469 consent for publication.

470 Availability of data and materials

471 The datasets used and/or analyzed during the current study are available from the
472 corresponding author upon reasonable request.

473 Competing interests

474 The authors report no biomedical financial interests or potential conflicts of interest.

475 Funding

476 This research was funded by the National Key Research and Development Program of
477 China (Nos. 2021YFC2500100 and 2021YFC2500103), the Beijing Municipal Natural
478 Science Foundation (No. 7244519), the National Natural Science Foundation of China
479 (Nos. 82071187, 81870821, 82172018, and 62201091), the Fundamental Research
480 Funds for the Central Universities (No. 2021XD-A03), the Science and Technology
481 Innovation 2030 Major Projects (No. 2022ZD0211600), and the Beijing Natural
482 Science Funds for Distinguished Young Scholars (No. JQ20036).

483 Authors' Contributions

484 JJ and ZK designed the study, performed analysis, and wrote the manuscript. LW
485 collected clinical and imaging data and performed analysis. ZP performed clustering
486 analysis. JS drew the figures. LW, RQ, ZM, WL, YS, JT, JS and ZH collected clinical,
487 imaging, and plasma data. WA, ZX, and KX performed statistical analysis. YH and HK
488 performed bioinformatic analysis. LY and DY checked the neuroimaging data. XJ and
489 LY designed the study, revised the manuscript, provided funding support, and
490 supervised all steps of the study.

491

492

493 References

- 494 1. Cummings J. (2021): The Role of Neuropsychiatric Symptoms in Research
495 Diagnostic Criteria for Neurodegenerative Diseases. *Am J Geriatr Psychiatry*
496 29:375-383.
- 497 2. Cerejeira J, Lagarto L, Mukaetova-Ladinska EB. (2012): Behavioral and
498 psychological symptoms of dementia. *Front Neurol* 3:73.
- 499 3. Aarsland D. (2020): Epidemiology and Pathophysiology of Dementia-Related
500 Psychosis. *J Clin Psychiatry* 81.
- 501 4. Marquié M, García-Gutiérrez F, Orellana A, Montreal L, de Rojas I,
502 García-González P, *et al.* (2023): The Synergic Effect of AT(N) Profiles and
503 Depression on the Risk of Conversion to Dementia in Patients with Mild
504 Cognitive Impairment. *Int J Mol Sci* 24.
- 505 5. Sanders AE. (2021): Cognition and Neuropsychiatric Symptoms in Alzheimer
506 Disease: A Century-Old Problem. *Neurology* 97:615-616.
- 507 6. Eikelboom WS, van den Berg E, Singleton EH, Baart SJ, Coesmans M, Leeuwis
508 AE, *et al.* (2021): Neuropsychiatric and Cognitive Symptoms Across the
509 Alzheimer Disease Clinical Spectrum: Cross-sectional and Longitudinal
510 Associations. *Neurology* 97:e1276-e1287.
- 511 7. Thiebaut de Schotten M, Forkel SJ. (2022): The emergent properties of the
512 connected brain. *Science* 378:505-510.
- 513 8. Innocenti GM, Schmidt K, Milleret C, Fabri M, Knyazeva MG, Battaglia-Mayer
514 A, *et al.* (2022): The functional characterization of callosal connections. *Prog*
515 *Neurobiol* 208:102186.
- 516 9. Axer M, Amunts K. (2022): Scale matters: The nested human connectome.
517 *Science* 378:500-504.
- 518 10. Vogel JW, Corriveau-Lecavalier N, Franzmeier N, Pereira JB, Brown JA, Maass
519 A, *et al.* (2023): Connectome-based modelling of neurodegenerative diseases:
520 towards precision medicine and mechanistic insight. *Nat Rev Neurosci*
521 24:620-639.
- 522 11. Pini L, Wennberg AM, Salvalaggio A, Vallesi A, Pievani M, Corbetta M. (2021):
523 Breakdown of specific functional brain networks in clinical variants of
524 Alzheimer's disease. *Ageing Res Rev* 72:101482.
- 525 12. Alexander-Bloch A, Giedd JN, Bullmore E. (2013): Imaging structural
526 co-variance between human brain regions. *Nat Rev Neurosci* 14:322-336.

- 527 13. Sebenius I, Seidlitz J, Warriier V, Bethlehem RAI, Alexander-Bloch A, Mallard
528 TT, *et al.* (2023): Robust estimation of cortical similarity networks from brain
529 MRI. *Nat Neurosci* 26:1461-1471.
- 530 14. Zhao K, Zheng Q, Che T, Dyrba M, Li Q, Ding Y, *et al.* (2021): Regional
531 radiomics similarity networks (R2SNs) in the human brain: Reproducibility,
532 small-world properties and a biological basis. *Netw Neurosci* 5:783-797.
- 533 15. Zhao K, Chen P, Alexander-Bloch A, Wei Y, Dyrba M, Yang F, *et al.* (2023): A
534 neuroimaging biomarker for Individual Brain-Related Abnormalities In
535 Neurodegeneration (IBRAIN): a cross-sectional study. *EClinicalMedicine*
536 65:102276.
- 537 16. Zhao K, Lin J, Dyrba M, Wang D, Che T, Wu H, *et al.* (2023): Coupling of the
538 spatial distributions between sMRI and PET reveals the progression of
539 Alzheimer's disease. *Netw Neurosci* 7:86-101.
- 540 17. Zhao K, Zheng Q, Dyrba M, Rittman T, Li A, Che T, *et al.* (2022): Regional
541 Radiomics Similarity Networks Reveal Distinct Subtypes and Abnormality
542 Patterns in Mild Cognitive Impairment. *Adv Sci (Weinh)* 9:e2104538.
- 543 18. Yu H, Ding Y, Wei Y, Dyrba M, Wang D, Kang X, *et al.* (2023): Morphological
544 connectivity differences in Alzheimer's disease correlate with gene
545 transcription and cell-type. *Hum Brain Mapp* 44:6364-6374.
- 546 19. Jiang J, Hong Y, Li W, Wang A, Jiang S, Jiang T, *et al.* (2023): Chain Mediation
547 Analysis of the Effects of Nutrition and Cognition on the Association of
548 Apolipoprotein E ϵ 4 with Neuropsychiatric Symptoms in Alzheimer's Disease.
549 *J Alzheimers Dis.*
- 550 20. Jack CR, Jr., Bennett DA, Blennow K, Carrillo MC, Dunn B, Haeberlein SB, *et al.*
551 (2018): NIA-AA Research Framework: Toward a biological definition of
552 Alzheimer's disease. *Alzheimers Dement* 14:535-562.
- 553 21. McKhann GM, Knopman DS, Chertkow H, Hyman BT, Jack CR, Jr., Kawas CH,
554 *et al.* (2011): The diagnosis of dementia due to Alzheimer's disease:
555 recommendations from the National Institute on Aging-Alzheimer's
556 Association workgroups on diagnostic guidelines for Alzheimer's disease.
557 *Alzheimers Dement* 7:263-269.
- 558 22. Dahnke R, Yotter RA, Gaser C. (2013): Cortical thickness and central surface
559 estimation. *Neuroimage* 65:336-348.
- 560 23. Fan L, Li H, Zhuo J, Zhang Y, Wang J, Chen L, *et al.* (2016): The Human
561 Brainnetome Atlas: A New Brain Atlas Based on Connectional Architecture.
562 *Cereb Cortex* 26:3508-3526.

- 563 24. Jiang J, Wang A, Shi H, Jiang S, Li W, Jiang T, *et al.* (2024): Clinical and
564 neuroimaging association between neuropsychiatric symptoms and
565 nutritional status across the Alzheimer's disease continuum: a longitudinal
566 cohort study. *J Nutr Health Aging* 28:100182.
- 567 25. Rolls ET, Huang CC, Lin CP, Feng J, Joliot M. (2020): Automated anatomical
568 labelling atlas 3. *Neuroimage* 206:116189.
- 569 26. Zhao K, Zheng Q, Dyrba M, Rittman T, Li A, Che T, *et al.* (2022): Regional
570 Radiomics Similarity Networks Reveal Distinct Subtypes and Abnormality
571 Patterns in Mild Cognitive Impairment. *Adv Sci (Weinh)* 9:e2104538.
- 572 27. Zhao K, Ding Y, Han Y, Fan Y, Alexander-Bloch AF, Han T, *et al.* (2020):
573 Independent and reproducible hippocampal radiomic biomarkers for
574 multisite Alzheimer's disease: diagnosis, longitudinal progress and biological
575 basis. *Sci Bull (Beijing)* 65:1103-1113.
- 576 28. Yang Z, Michailidis G. (2016): A non-negative matrix factorization method for
577 detecting modules in heterogeneous omics multi-modal data. *Bioinformatics*
578 32:1-8.
- 579 29. Chen P, Yao H, Tijms BM, Wang P, Wang D, Song C, *et al.* (2023): Four Distinct
580 Subtypes of Alzheimer's Disease Based on Resting-State Connectivity
581 Biomarkers. *Biological Psychiatry* 93:759-769.
- 582 30. Hawrylycz MJ, Lein ES, Guillozet-Bongaarts AL, Shen EH, Ng L, Miller JA, *et al.*
583 (2012): An anatomically comprehensive atlas of the adult human brain
584 transcriptome. *Nature* 489:391-399.
- 585 31. Markello RD, Arnatkeviciute A, Poline JB, Fulcher BD, Fornito A, Misic B.
586 (2021): Standardizing workflows in imaging transcriptomics with the abagen
587 toolbox. *Elife* 10.
- 588 32. Whitaker KJ, Vertes PE, Romero-Garcia R, Vasa F, Moutoussis M, Prabhu G, *et al.*
589 (2016): Adolescence is associated with genomically patterned
590 consolidation of the hubs of the human brain connectome. *Proc Natl Acad Sci*
591 *U S A* 113:9105-9110.
- 592 33. Zhou Y, Zhou B, Pache L, Chang M, Khodabakhshi AH, Tanaseichuk O, *et al.*
593 (2019): Metascape provides a biologist-oriented resource for the analysis of
594 systems-level datasets. *Nat Commun* 10:1523.
- 595 34. Seguin C, Sporns O, Zalesky A. (2023): Brain network communication:
596 concepts, models and applications. *Nat Rev Neurosci* 24:557-574.
- 597 35. Huang L, Huang Q, Xie F, Guo Q. (2024): Neuropsychiatric symptoms in
598 Alzheimer's continuum and their association with plasma biomarkers. *J Affect*
599 *Disord* 348:200-206.

- 600 36. Ma H, Lu X, Zhou A, Wang F, Zuo X, Zhan M, *et al.* (2022): Clinical Practice
601 Guidelines for the Management of Behavioral and Psychological Symptoms of
602 Dementia: A Systematic Review With AGREE II. *Front Neurol* 13:799723.
- 603 37. Serra L, Bruschini M, Di Domenico C, Mancini M, Bechi Gabrielli G, Bonarota
604 S, *et al.* (2020): Behavioral psychological symptoms of dementia and
605 functional connectivity changes: a network-based study. *Neurobiol Aging*
606 94:196-206.
- 607 38. Chatterjee P, Pedrini S, Doecke JD, Thota R, Villemagne VL, Doré V, *et al.*
608 (2023): Plasma A β 42/40 ratio, p-tau181, GFAP, and NfL across the
609 Alzheimer's disease continuum: A cross-sectional and longitudinal study in
610 the AIBL cohort. *Alzheimers Dement* 19:1117-1134.
- 611 39. Smirnov DS, Ashton NJ, Blennow K, Zetterberg H, Simrén J, Lantero-Rodriguez
612 J, *et al.* (2022): Plasma biomarkers for Alzheimer's Disease in relation to
613 neuropathology and cognitive change. *Acta Neuropathol* 143:487-503.
- 614 40. Kwak S, Park S, Kim J, Park S, Lee JY. (2020): Multivariate neuroanatomical
615 correlates of behavioral and psychological symptoms in dementia and the
616 moderating role of education. *Neuroimage Clin* 28:102452.
- 617 41. Jiang J, Wang A, Liu Y, Yao Z, Sun M, Jiang T, *et al.* (2023): Spatiotemporal
618 Characteristics of Regional Brain Perfusion Associated with Neuropsychiatric
619 Symptoms in Patients with Alzheimer's Disease. *J Alzheimers Dis* 95:981-993.
- 620 42. Novikova G, Kapoor M, Tcw J, Abud EM, Efthymiou AG, Chen SX, *et al.* (2021):
621 Integration of Alzheimer's disease genetics and myeloid genomics identifies
622 disease risk regulatory elements and genes. *Nat Commun* 12:1610.
- 623 43. Huang MF, Lee WJ, Yeh YC, Liao YC, Wang SJ, Yang YH, *et al.* (2020): Genetics
624 of neuropsychiatric symptoms in patients with Alzheimer's disease: A 1-year
625 follow-up study. *Psychiatry Clin Neurosci* 74:645-651.
- 626 44. Hartmann K, Sadée CY, Satwah I, Carrillo-Perez F, Gevaert O. (2023): Imaging
627 genomics: data fusion in uncovering disease heritability. *Trends Mol Med*
628 29:141-151.
- 629 45. Sun Y, Lin J, Zhang L. (2019): The application of weighted gene co-expression
630 network analysis in identifying key modules and hub genes associated with
631 disease status in Alzheimer's disease. *Ann Transl Med* 7:800.
- 632 46. Shah D, Gsell W, Wahis J, Luckett ES, Jamouille T, Vermaercke B, *et al.* (2022):
633 Astrocyte calcium dysfunction causes early network hyperactivity in
634 Alzheimer's disease. *Cell Rep* 40:111280.
- 635 47. Garrido JJ. (2023): Contribution of Axon Initial Segment Structure and
636 Channels to Brain Pathology. *Cells* 12.

- 637 48. Channer B, Matt SM, Nickoloff-Bybel EA, Pappa V, Agarwal Y, Wickman J, *et*
638 *al.* (2023): Dopamine, Immunity, and Disease. *Pharmacol Rev* 75:62-158.
- 639 49. Pourhamzeh M, Moravej FG, Arabi M, Shahriari E, Mehrabi S, Ward R, *et al.*
640 (2022): The Roles of Serotonin in Neuropsychiatric Disorders. *Cell Mol*
641 *Neurobiol* 42:1671-1692.
- 642 50. Schmidt SG, Malle MG, Nielsen AK, Bohr SS, Pugh CF, Nielsen JC, *et al.* (2022):
643 The dopamine transporter antiports potassium to increase the uptake of
644 dopamine. *Nat Commun* 13:2446.
- 645 51. Huang J, Xu Q, Li Y, He X, Guo Y, Sun X. (2022): Activation of Calcium-Sensing
646 Receptor in the Area Postrema Inhibits Food Intake via Glutamatergic and
647 GABAergic Signaling Pathways. *Mol Nutr Food Res* 66:e2200245.

648

649

650

651 Figure Legends

652 Fig.1 Schematic of the data analysis pipeline

653 The data were preprocessed, and the intensity and textural features were extracted
 654 using the Brainnetome Atlas. Next, the network was achieved by computing the
 655 Pearson correlation coefficient of each pairwise brain region. Finally, the clinical and
 656 connective differences were compared among the NC, AD-nNPS, and AD-NPS groups
 657 (a). Clustering the NPS group into three different subtypes with NPS, including sNPS,
 658 miNPS, and moNPS, based on the differential connection between AD-NPS and
 659 AD-nNPS groups (b). The clinical presentation abnormalities, multimodal
 660 neuroimaging measures, gene enrichment profiles and GWAS, and longitudinal
 661 progression were compared between each of the three NPS subtypes and AD-nNPS,
 662 respectively (c).

663 *Abbreviations: R2SN, regional radiomics similarity network; NC, normal control; AD,*
 664 *Alzheimer's disease; NPS, neuropsychiatric symptoms; nNPS, patients with*
 665 *Alzheimer's disease without NPS; NMF, nonnegative matrix factorization; sNPS,*
 666 *severe neuropsychiatric symptoms; miNPS, mild neuropsychiatric symptoms; moNPS,*
 667 *moderate neuropsychiatric symptoms; GWAS, genome-wide association studies*

668

669 Fig.2 Characteristics of altered morphological connectivity in the R2SN profiles

670 Compared with the NCs, AD patients with NPSs (AD-NPS group) showed alterations
 671 in the R2SN in the psychobehavioral-related brain areas (a), while the R2SN
 672 alterations in those without NPSs (AD-nNPS group) on the AD continuum were
 673 concentrated in the cognition-related brain areas (b). Three hundred key
 674 connections between patients with or without NPSs in the AD continuum were used
 675 for clustering analysis (c). Compared to patients with nNPS, patients with sNPS and
 676 moNPS showed a significant opposing damage pattern in the R2SN in the

medioventral occipital cortex, inferior temporal gyrus, fusiform gyrus, superior parietal lobule, and thalamus (d, f), while lesser alternations were observed in those with miNPS (e). Differences in the R2SN in sNPS versus miNPS (g), moNPS versus miNPS (h), and sNPS versus moNPS (i). “Red” indicates increased connections, and “blue” indicates decreased connections.

Abbreviations: R2SN, regional radiomics similarity network; NCs, normal controls; AD, Alzheimer’s disease; NPS, neuropsychiatric symptoms; nNPS, patients with Alzheimer’s disease without NPS; NMF, nonnegative matrix factorization; sNPS, severe neuropsychiatric symptoms; miNPS, mild neuropsychiatric symptoms; moNPS, moderate neuropsychiatric symptoms.

Fig.3 Differences in multimodal neuroimaging burden between three NPS subtypes and nNPS in the AD continuum

The differences in the regional GM volume (a-c), CT (d-f), and regional cCBF values (g-i) in sNPS versus nNPS, miNPS versus nNPS, and moNPS versus nNPS. The color bar indicates the *T*-value. “Red” indicates increased *T*-value, and “blue” indicates decreased *T*-value.

Abbreviations: NPS, neuropsychiatric symptoms; AD, Alzheimer’s disease; nNPS, patients with Alzheimer’s disease without NPSs; sNPS, severe neuropsychiatric symptoms; miNPS, mild neuropsychiatric symptoms; moNPS, moderate neuropsychiatric symptoms; GM, grey matter; CT, cortical thickness; cCBF, corrected cerebral blood flow

Fig.4 GSEA between three NPS subtypes and AD-nNPS in the AD continuum

The weighted regional gene expression and the correlation of the PLS1 score and the *T*-map of the difference in sNPS versus nNPS (a), miNPS versus nNPS (b), and moNPS

versus nNPS (c). GSEA revealed the top 20 significant typical Gene Ontology terms and KEGG pathways summarizing the functional role of the 15 633 genes ranked according to PLS weight in the sNPS (d), miNPS (e), and moNPS (f) groups. Inorganic ion transmembrane transport (sodium, potassium) and synaptic signaling (catecholamine, dopamine) were correlated with sNPS, while inorganic ion transmembrane transport (calcium, anion) and synaptic signaling (glutamate, GABA) were correlated with moNPS.

Abbreviations: GSEA, gene set enrichment analysis; NPS, neuropsychiatric symptoms; sNPS, severe neuropsychiatric symptoms; miNPS, mild neuropsychiatric symptoms; moNPS, moderate neuropsychiatric symptoms; KEGG, Kyoto Encyclopedia of Genes and Genomes; PLS, partial least squares; nNPS, patients with Alzheimer's disease without NPSs; sNPS, severe neuropsychiatric symptoms; miNPS, mild neuropsychiatric symptoms; moNPS, moderate neuropsychiatric symptoms

Fig.5 Genome-wide association studies of AD patients with specific NPS subtype

Two genomic risk loci significantly associated with sNPS (rs16879809 and rs11656034), with the top SNPs in these regions mapping to 13 genes, which are enriched in the pathway related to logical memory in mild cognitive impairment (A). Five genomic risk loci significantly associated with miNPS, with the top SNPs in these regions mapping to 38 genes enriched in the diabetic retinopathy in the type 2 diabetes pathway (B). Five genomic risk loci significantly associated with moNPS, where the top SNPs in these regions mapped to 29 genes associated with the pathway for non-response to antidepressants and depression (C).

729 **Tables**

730 Table 1. Demographic, clinical, and biomarker characteristics of all participants at

731 baseline

732

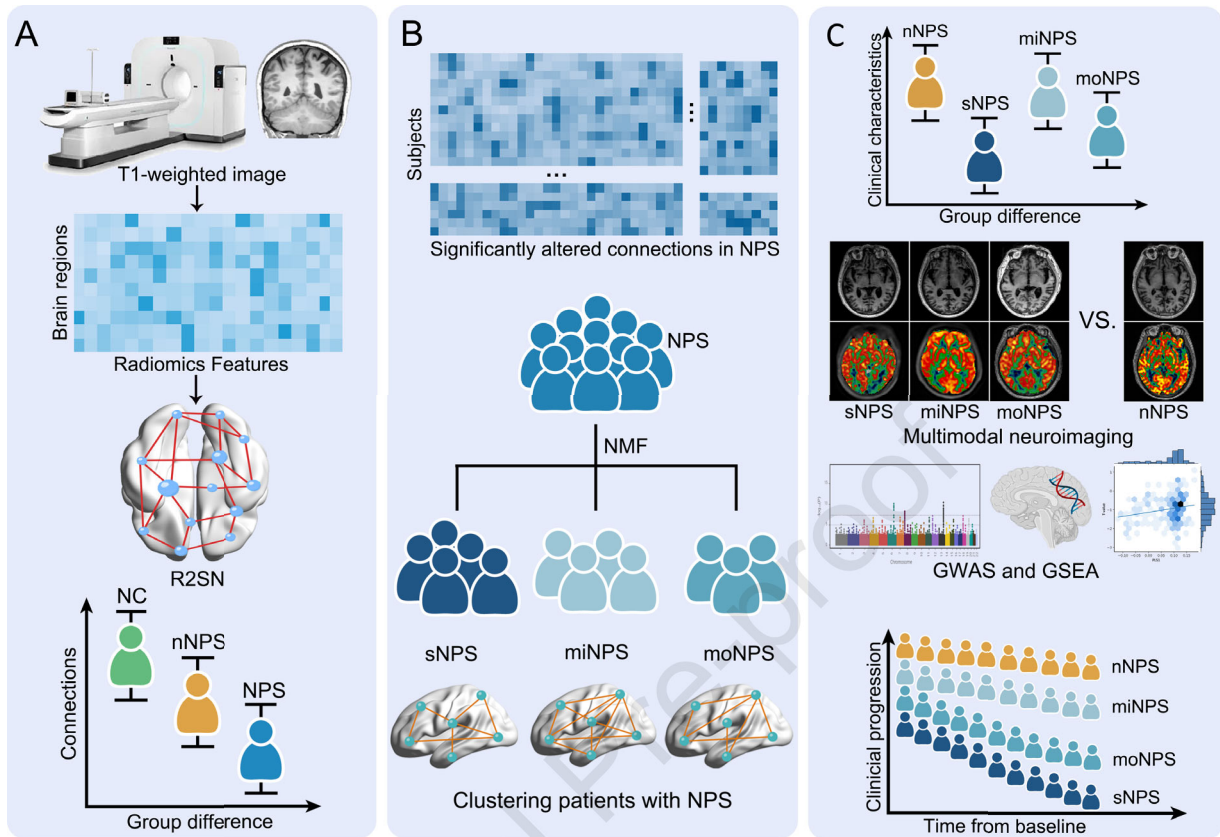
Tables

Table 1. Demographic, clinical, and biomarker characteristics of all participants at baseline

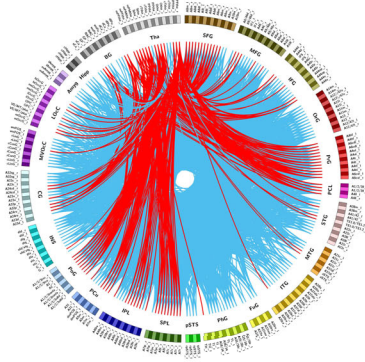
Variables	NCs (n = 63)	AD-nNPS (n = 111)	AD-NPS (n = 376)	F/ χ^2 /K	P value ^a	P value ^b
Demographics						
Age (years, mean $\pm$ SD)	61.43 $\pm$ 8.13	66.61 $\pm$ 7.45	67.57 $\pm$ 8.89	13.983	< 0.001	0.591
Sex [female (%)]	49.00 (77.78)	70.00 (63.06)	226.00 (60.11)	7.214	0.027	0.575
Education [years, median (IQR)]	14.00 (11.00, 14.00)	11.00 (8.00, 13.00)	11.00 (8.00, 12.00)	29.143	< 0.001	0.769
APOE ϵ 4 carrier (%)	8.00 (12.70)	32.00 (28.83)	163.00 (43.35)	7.963	0.019	0.607
Marital status (yes, %)	57.00 (90.48)	104.00 (93.69)	327.00 (86.97)	4.093	0.129	0.051
Medical histories						
Hypertension (yes, %)	15.00 (23.81)	44.00 (39.64)	163.00 (43.35)	8.590	0.014	0.487
Diabetes mellitus (yes, %)	7.00 (11.11)	18.00 (16.22)	70.00 (18.62)	2.236	0.327	0.564
Cerebrovascular disease (yes, %)	3.00 (4.76)	14.00 (12.61)	71.00 (18.88)	9.193	0.010	0.126
Coronary heart disease (yes, %)	9.00 (14.29)	19.00 (17.12)	73.00 (19.41)	1.091	0.580	0.587
Dyslipidemia (yes, %)	22.00 (34.92)	50.00 (45.05)	158.00 (42.02)	1.713	0.425	0.571
Smoking (yes, %)	9.00 (14.26)	23.00 (20.72)	77.00 (20.48)	1.374	0.503	0.956
Alcohol consumption (yes, %)	18.00 (28.57)	28.00 (25.23)	102.00 (27.13)	0.258	0.879	0.691
Anesthesia (yes, %)	39.00 (61.90)	60.00 (54.05)	195.00 (51.86)	2.208	0.332	0.684
Plasma pathological biomarkers (N=135)						
p-Tau181 [pg/ml, median (IQR)]	-	1.01 (0.83, 1.90)	1.38 (0.96, 2.60)	-1.775		0.076
t-Tau [pg/ml, median (IQR)]	-	17.45 (10.32, 22.28)	13.67 (9.59, 20.30)	-0.955		0.340
A β ₄₂ [pg/ml, median (IQR)]	-	15.54 (12.87, 18.79)	15.63 (12.90, 19.04)	-0.102		0.919
A β ₄₀ [pg/ml, median (IQR)]	-	107.02 (99.25, 130.64)	114.74 (99.28, 130.20)	-0.553		0.581
A β _{42/40} [median (IQR)]	-	0.15 (0.10, 0.19)	0.14 (0.12, 0.16)	-0.841		0.400
Neuropsychological battery						
MMSE [score, median (IQR)]	28.00 (27.00, 28.00)	26.00 (22.50, 28.00)	23.00 (16.00, 26.00)	120.616	< 0.001	< 0.001
MoCA [score, median (IQR)]	27.00 (26.00, 27.00)	21.00 (18.00, 24.00)	17.50 (11.00, 22.00)	170.406	< 0.001	< 0.001
ADL [score, median (IQR)]	20.00 (20.00, 20.00)	20.00 (20.00, 20.00)	22.00 (20.00, 31.00)	114.780	< 0.001	< 0.001
MNA [score, median (IQR)]	26.00 (24.00, 26.00)	25.50 (23.50, 27.00)	23.00 (20.50, 25.00)	50.244	< 0.001	< 0.001
CBI [score, median (IQR)]	-	0.00 (0.00, 0.50)	9.00 (0.00, 28.00)	-	-	< 0.001
PSQI [score, median (IQR)]	5.00 (3.00, 8.00)	5.00 (2.00, 6.00)	5.00 (2.00, 9.25)	12.348	0.002	0.001

^a R $\times$ C diagram χ^2 test, univariate analysis of variance or Kruskal–Wallis test were used, and $P < 0.05$ was set as significant; ^b Pairwise comparisons between patients with and without NPSs using the independent t test, Mann–Whitney U test, or χ^2 test ($P < 0.025$ indicated statistical significance).

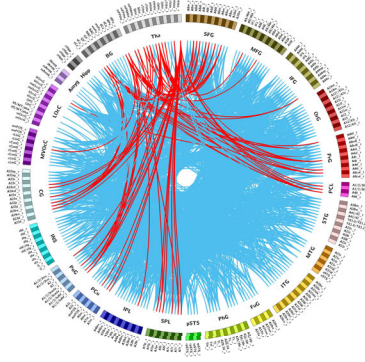
NCs, normal controls; NPS, neuropsychiatric symptom; AD-NPS, patients on the AD continuum with NPS; AD-nNPS, patients on the AD continuum without NPSs; APOE ϵ 4, apolipoprotein E type epsilon 4; p-Tau₁₈₁, tau phosphorylated at amino acid 181; t-Tau, total tau; A β ₄₂, the 42 amino acid-long amyloid- β ; A β ₄₀, the 40 amino acid-long amyloid- β ; A β _{42/40}, the ratio of A β ₄₂ to A β ₄₀; MMSE, Mini-Mental State Examination; MoCA, Montreal Cognitive Assessment; NPI, Neuropsychiatric Inventory; ADL, Activities of Daily Living; MNA, Mini-Nutritional Assessment; CBI, Caregiver Burden Inventory; PSQI, Pittsburgh sleep quality index; IQR, interquartile range.



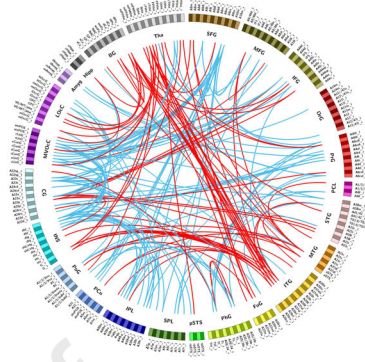
A NPS vs. NC



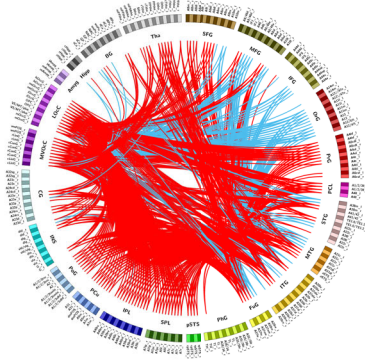
B nNPS vs. NC



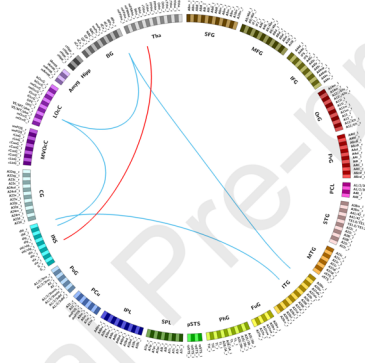
C NPS vs. nNPS



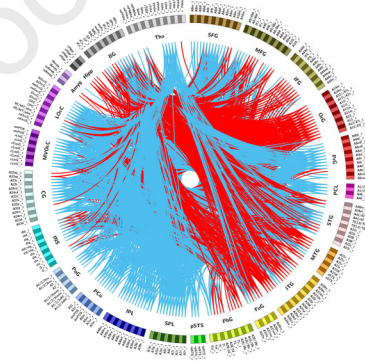
D sNPS vs. nNPS



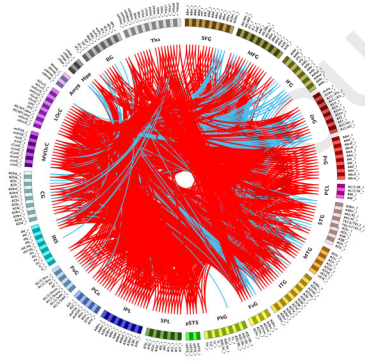
E miNPS vs. nNPS



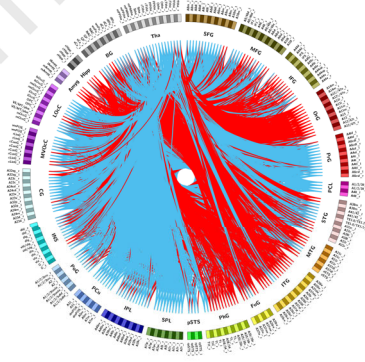
F moNPS vs. nNPS



G sNPS vs. miNPS



H moNPS vs. miNPS



I sNPS vs. moNPS

